AFFINENESS OF THE LIMIT VIA INTEGRAL APPROXIMATION

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Affineness lemma. Limits, Lemma lemma-affine (same chapter)
- Integral ring maps. Algebra, integral ring maps

Source reference. Original reference 01WM: sources/stacks-chow/tags.txt, source lines 2487–2487. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09MQ: sources/stacks-chow/tags.txt, source lines 12188–12188. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SG: sources/stacks-chow/tags.txt, source lines 2337–2337. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09YZ: sources/stacks-chow/tags.txt, source lines 12582–12582. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z6: sources/stacks-chow/tags.txt, source lines 2581–2581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZT: sources/stacks-chow/tags.txt, source lines 2606–2606. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Let $f : X \to S$ be a morphism of schemes. Assume X is affine and that f is surjective and universally closed[1]*. Then S is affine.*

Proof. By Morphisms, Lemma 09MQ the scheme S is separated. Then by Morphisms, Lemma 01SG we find that f is affine. Whereupon by Morphisms, Lemma 01WM we see that f is integral.

By the preceding paragraph, we may assume $f : X \to S$ is surjective and integral, X is affine, and S is separated. Since f is surjective and X is quasi-compact we also deduce that S is quasi-compact.

By Lemma 09YZ we can write $X = \lim_i X_i$ with $X_i \to S$ finite. By Lemma 01Z6 we see that for i sufficiently large the scheme X_i is affine. Moreover, since $X \to S$ factors through each X_i we see that $X_i \to S$ is surjective. Hence we conclude that S is affine by Lemma 01ZT. □

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[1] An integral morphism is universally closed, see Morphisms, Lemma 01WM.